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(54) **TRANSFORMABLE FOLDABLE PAPER BAG**

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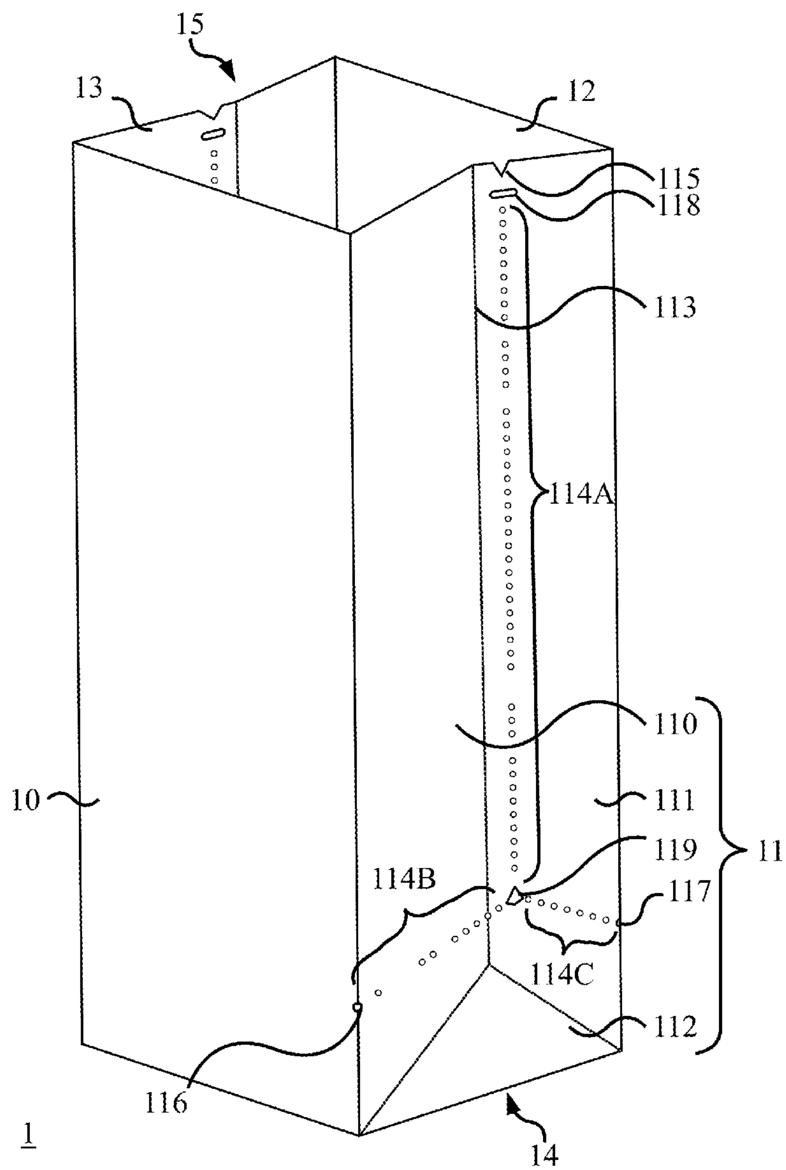
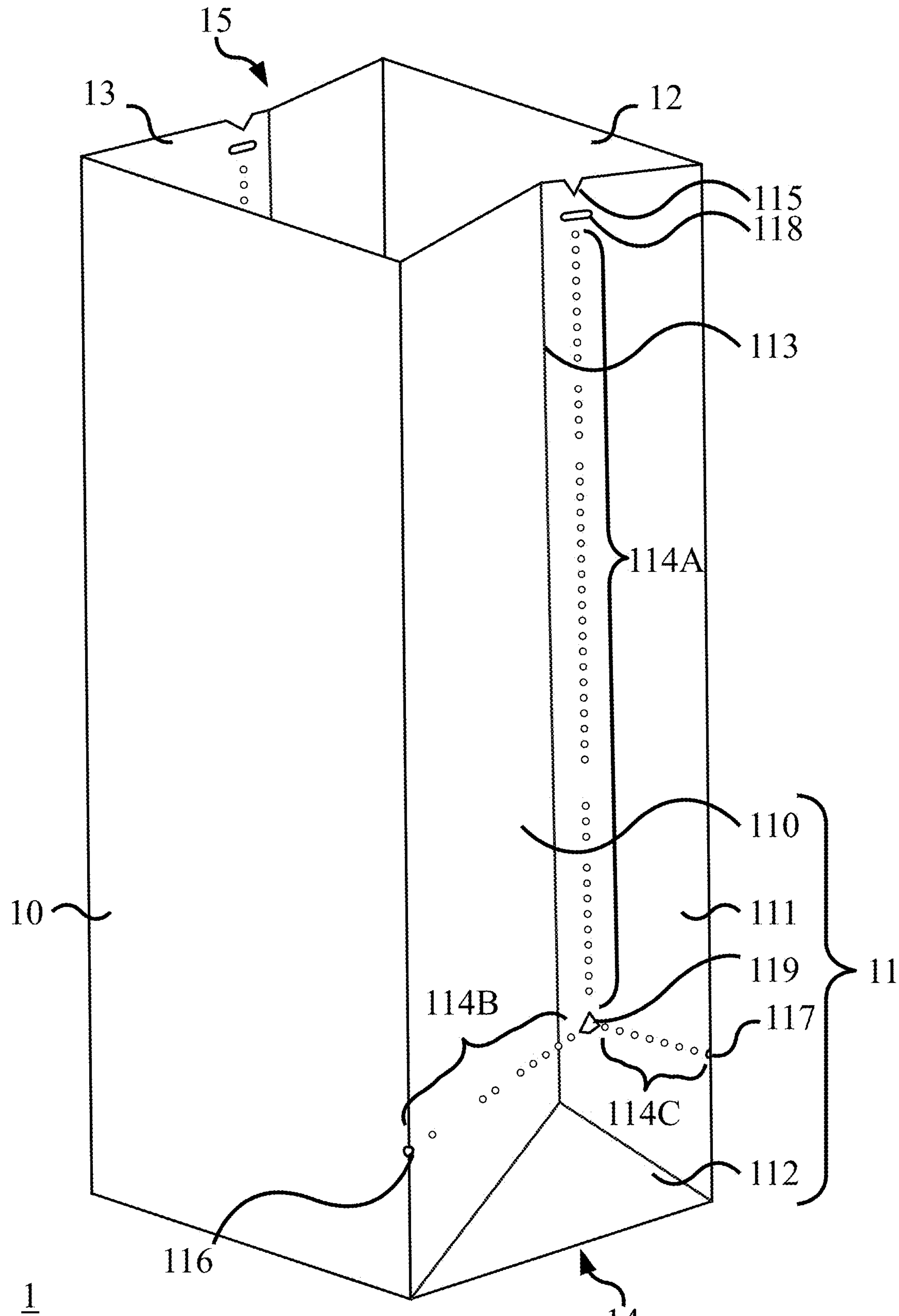
(57) **ABSTRACT**

A foldable paper bag is provided. The foldable paper bag has two perforated panels being parallel. Every perforated panel has perforations arranged in a vertically flipped "Y" shape. When the foldable paper bag is in an upright position, every arm of the vertically flipped "Y" shape is connected to one of the vertical edges of the perforated panel, and the leg of the vertically flipped "Y" shape is vertically extend towards an opening of the foldable paper bag. The provided foldable paper bag can be used as a clean, tidy, and disposable transformable container for food and dining utensil.




15
13
12
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114A
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FIG. 1

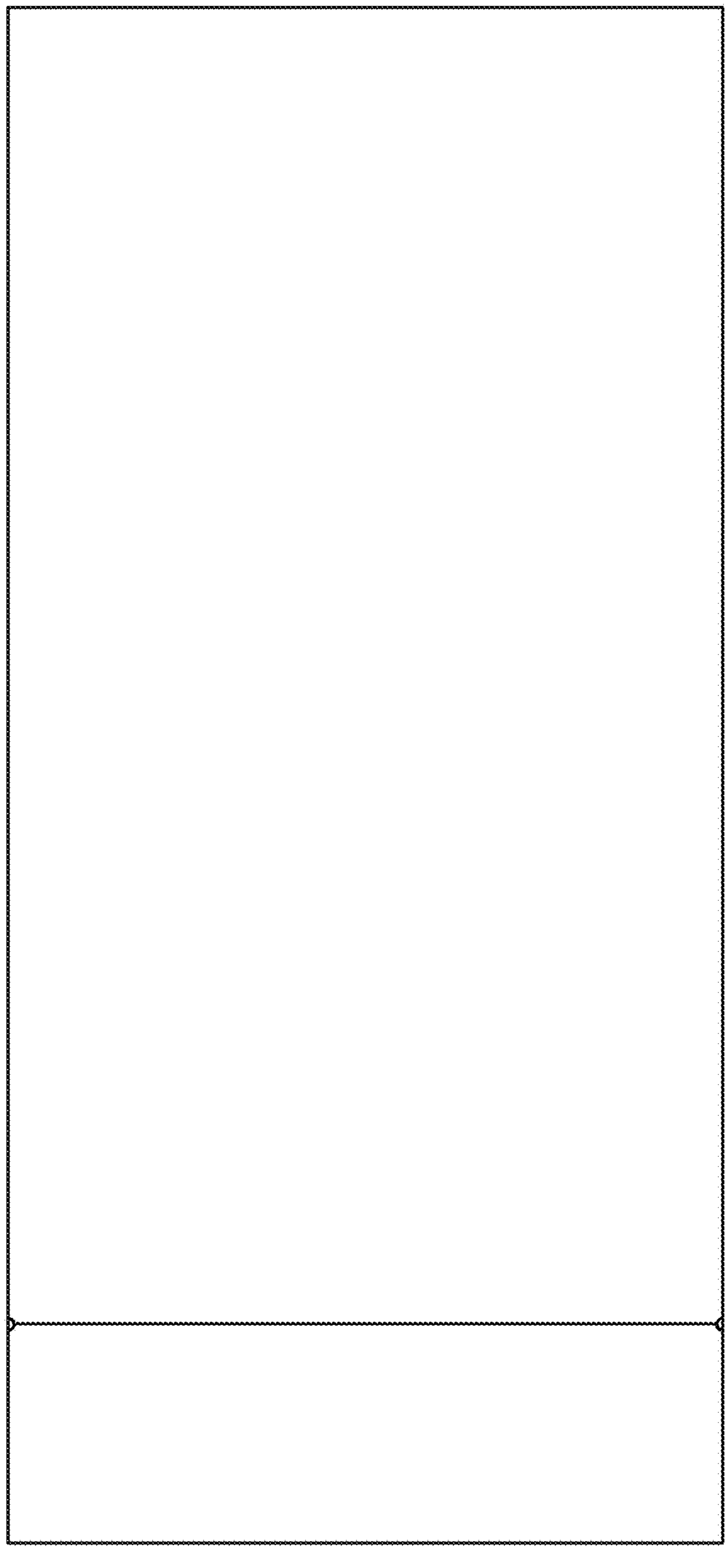

FIG. 2

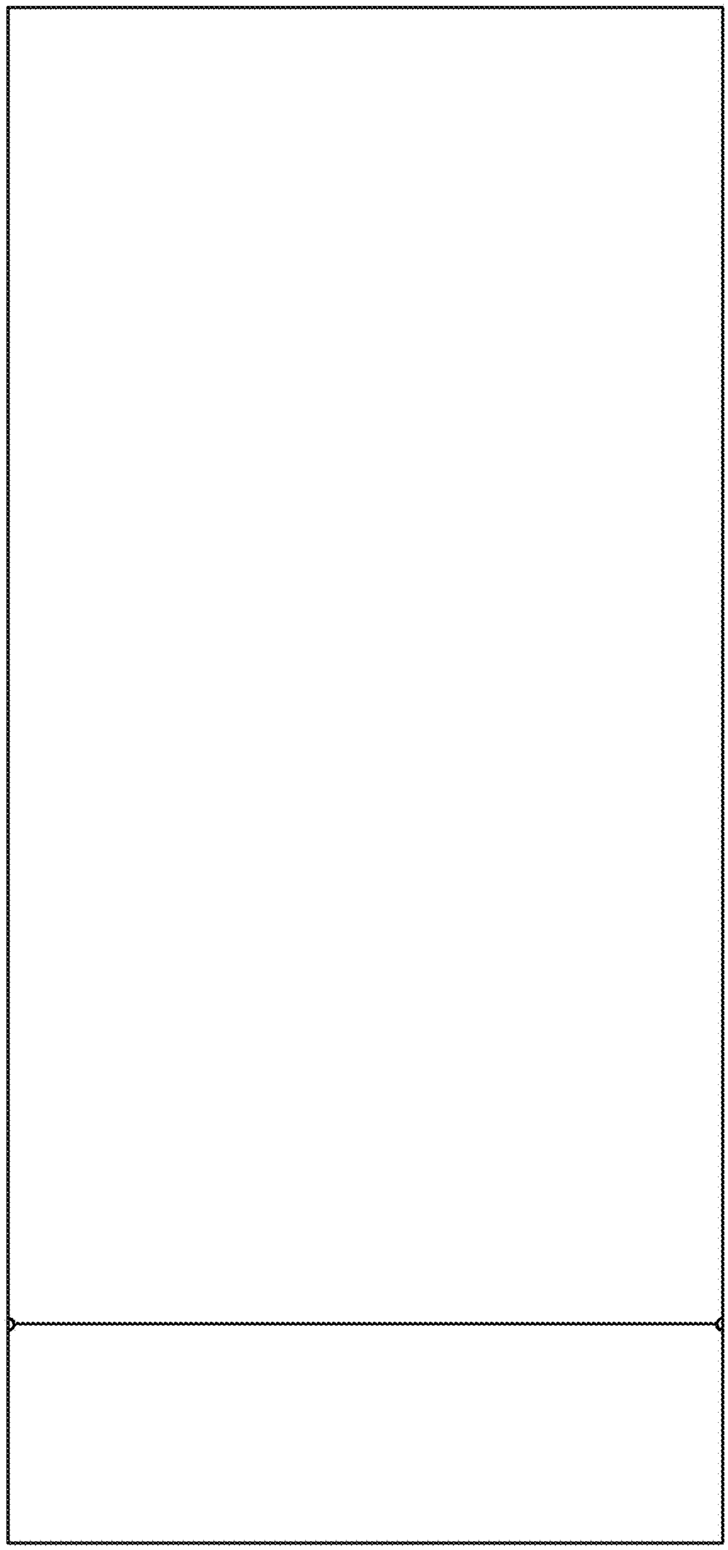

FIG. 3

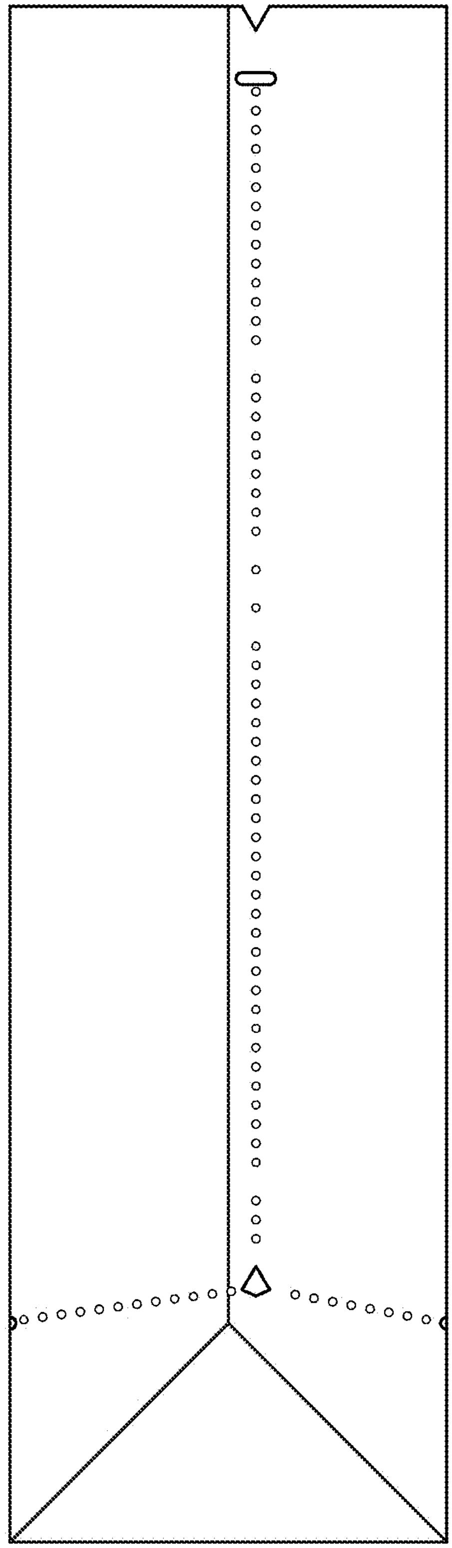

FIG. 4

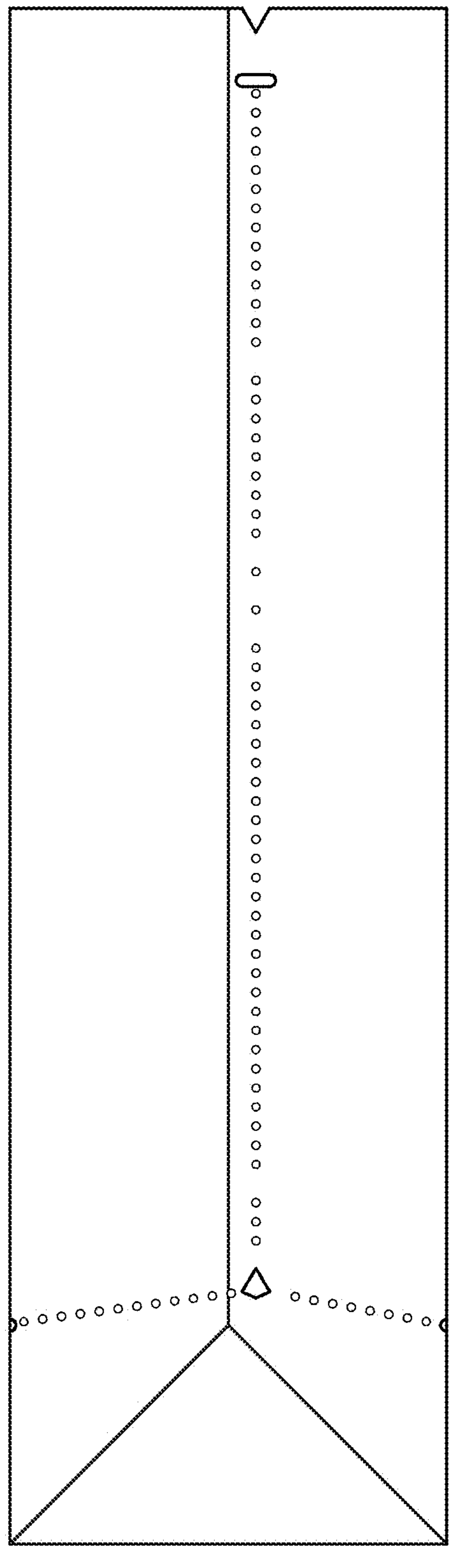

FIG. 5

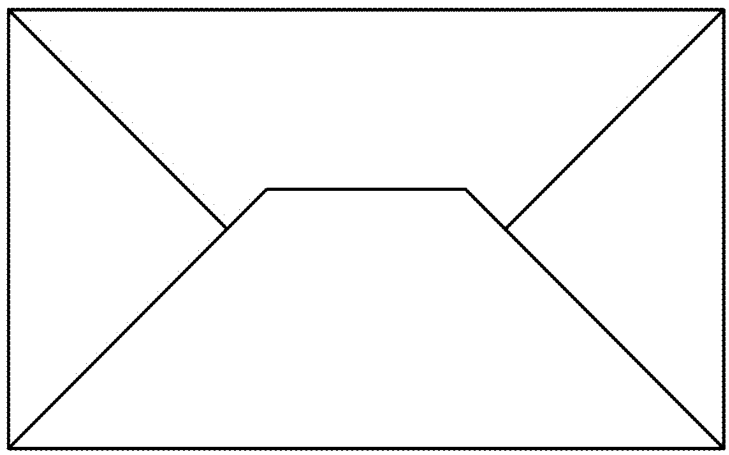

FIG. 6

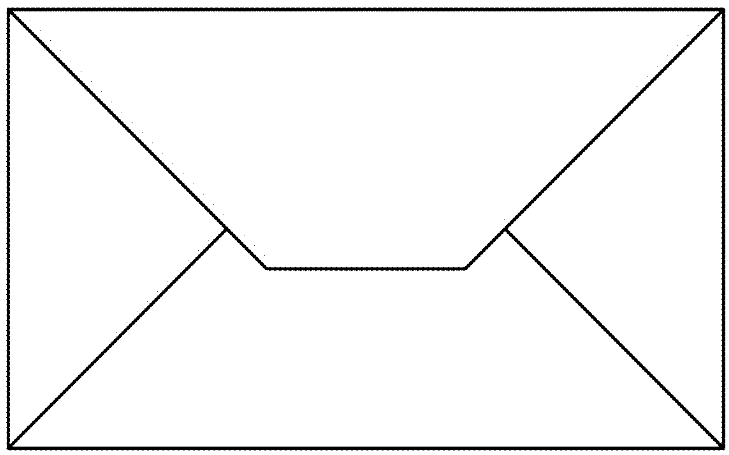

FIG. 7

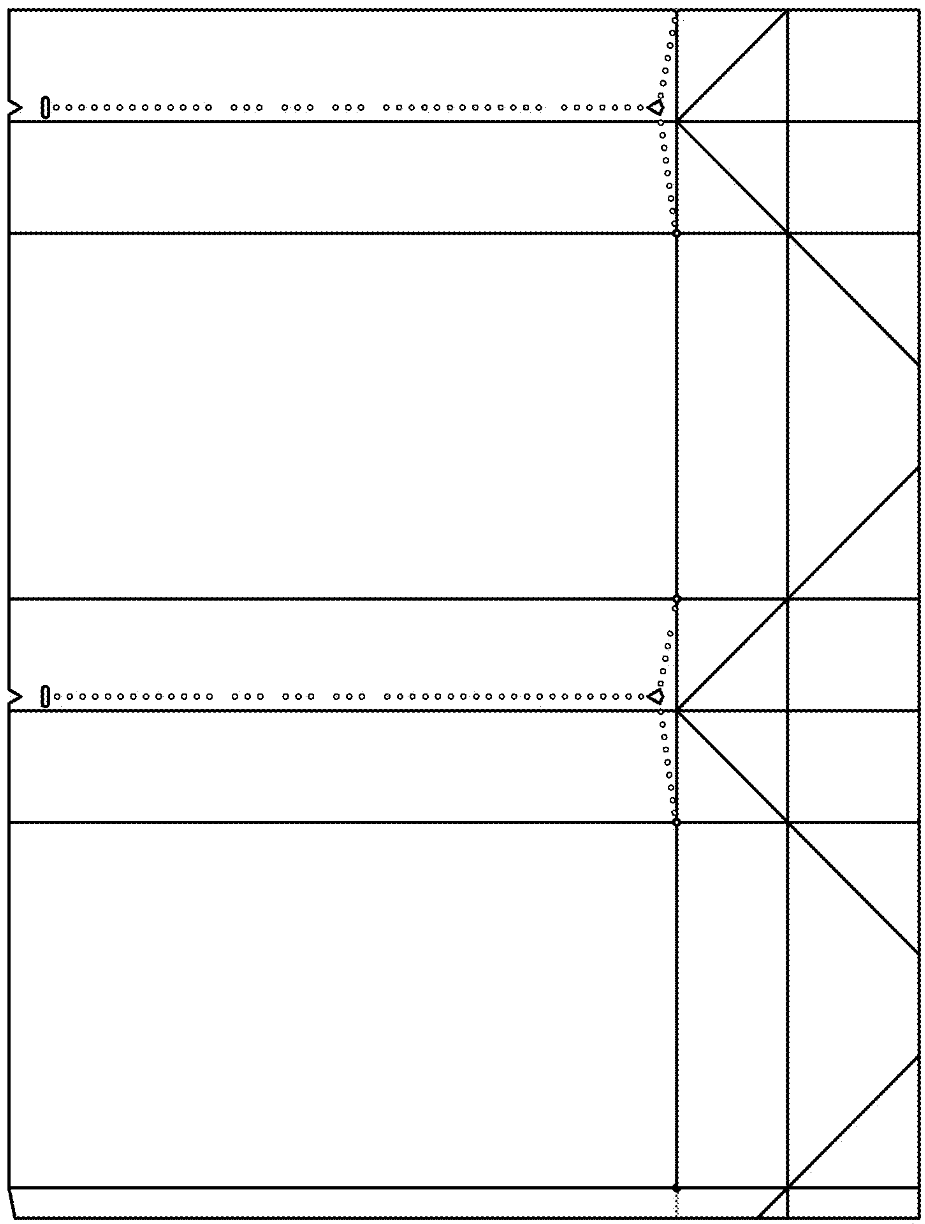

FIG. 8

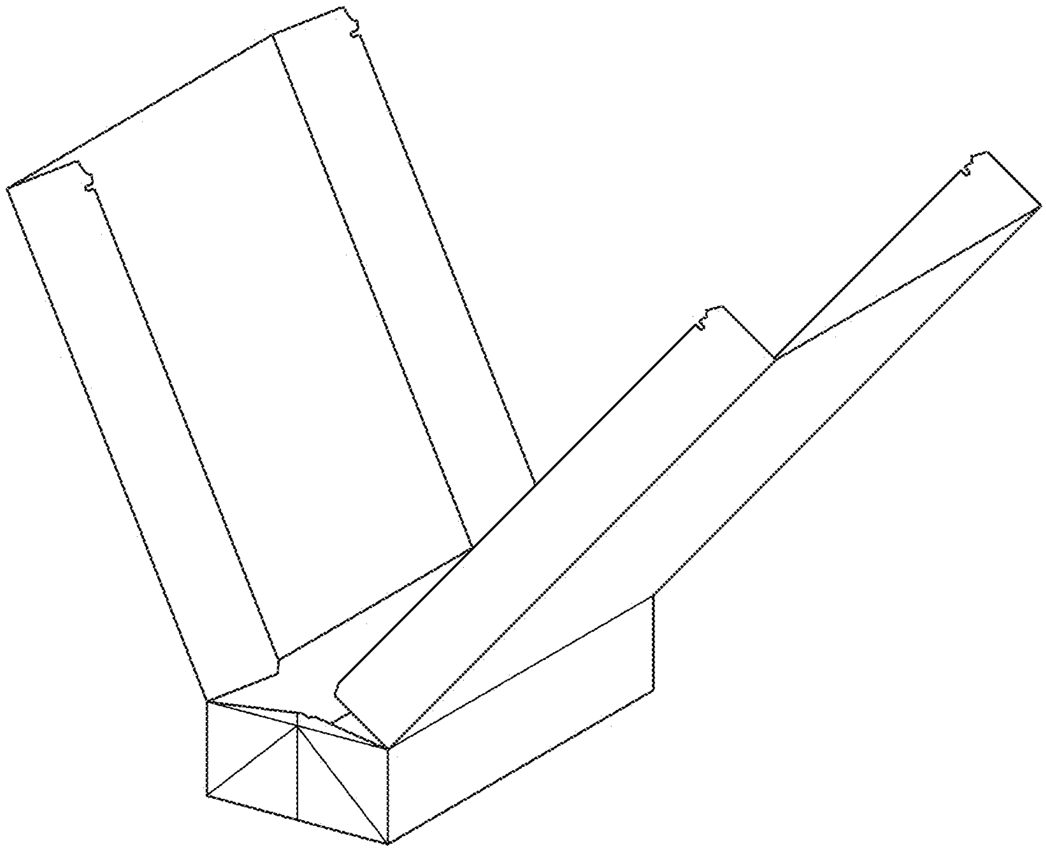

FIG. 9

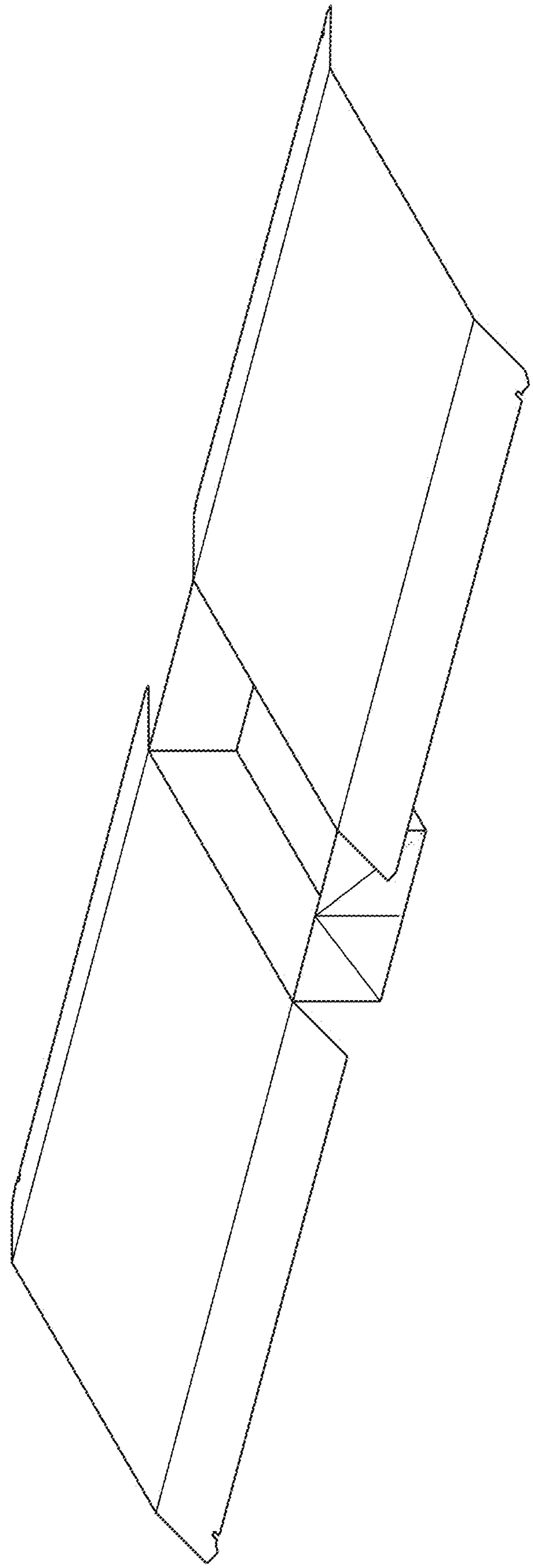

FIG. 10

TRANSFORMABLE FOLDABLE PAPER BAG

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority from the U.S. Provisional Patent Application No. 63/562,269 filed on Mar. 7, 2024, and the disclosure of which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention pertains to the foldable and transforming paper bags.

BACKGROUND OF THE INVENTION

[0003] The prevalence of paper bags (a.k.a. brown bags) has become a ubiquitous item in shops and restaurants as a means of packaging and carrying food and goods. In the fast-food industry, these paper bags are extensively utilized for packaging meals for customers due to their cost-effectiveness and environmental friendliness. However, a significant challenge arises when consumers wish to consume their food directly out of these packaging. The limited surface area provided by the paper bags and even along with napkins included within them often proves inadequate for a comfortable setting and enjoyment of a sit-down meal. The traditional paper bag makes a very poor dining utensil. The napkins also have a tendency of being easily soaked.

[0004] Therefore, there is a need for innovative solutions that enhance the utility of the foldable paper bags in the context of fast-food consumption. Specifically, there is an opportunity to design a foldable paper bag that can be transformed into a proper dining utensil for consumers to place their food therewithin and eat out of. Such a design could revolutionize the way people carry and enjoy fast food on the go, providing a clean, tidy, and disposable transformable food container-dining utensil. This innovative approach would not only improve the user experience but also contribute to the overall sustainability and efficiency of fast-food packaging.

SUMMARY OF THE INVENTION

[0005] It is an objective of the present invention to address the shortcomings in the state of the art by providing a clean, tidy, and disposable transformable container for food and dining utensil.

[0006] In accordance with a first aspect of the present invention, a foldable paper bag that can be transformed is provided. The foldable paper bag includes two flat panels, two perforated panels, and a bottom panel, and each of the perforated panels connects the flat panels, and the bottom panel connects the flat panels and the perforated panels. The flat panels are parallel, and the perforated panels are parallel. The flat panels, the perforated panels, and the bottom panel form a tubular cavity. Each of the perforated panels has a first series of perforations, a second series of perforations, and a third series of perforations. The first series of perforations are arranged along a first straight line, which is parallel with a normal of a surface of the bottom panel in a carrying mode. The second series of perforations are arranged along a second straight line, and the third series of perforations are arranged along a third straight line. An end of the first straight line is adjacent to an opening of the foldable paper bag, and the other end of the first straight line is adjacent to an end of the second straight line and an end of the third straight line. Another end of the second straight line is connected to a side connecting the perforated panel and one of the flat panels, and another end of the third straight line is connected to another side connecting the perforated panel and another flat panel.

[0007] In accordance with a second aspect of the present invention, a foldable paper bag that can be transformed is provided. The foldable paper bag has two perforated panels that are parallel, and every perforated panel has perforations arranged in a vertically flipped "Y" shape. When the foldable paper bag is in an upright position, every arm of the vertically flipped "Y" shape is connected to one of the vertical edges of the perforated panel, and the leg of the vertically flipped "Y" shape is vertically extend towards the opening of the foldable paper bag.

[0008] In accordance with a third aspect of the present invention, a foldable paper bag that can be transformed is provided. The foldable paper bag has two perforated panels that are parallel, and two flat panels that are parallel. Every perforated panel has perforations that divided the perforated panel into a front section, a back section, and a lower section. When the foldable paper bag is in an upright position, the front section is connected to the front flat panel, and the back section is connected to the back flat panel, and the lower section and a bottom panel of the foldable paper bag are connected.

[0009] In accordance with one embodiment of the present invention, every perforated panel has a crease, and the crease extends vertically when the foldable paper bag is in an upright position, and the crease and the portion of the perforations that form a vertical straight line keep a gap in-between.

[0010] In accordance with one embodiment of the present invention, every perforated panel has a notch on the edge that form the opening of the foldable paper bag, and the notch is aligned with a portion of the perforations that is vertically arranged when the foldable paper bag is in an upright position.

[0011] In accordance with another embodiment of the present invention, the foldable paper bag is made of paper, and the orientation of the paper is grain long.

[0012] In accordance with another embodiment of the present invention, the perforated panel has a hole form on the vertical edge, and the hole is aligned with portion of the perforations that is not vertically arranged when the foldable paper bag is in an upright position.

[0013] In accordance with another embodiment of the present invention, the foldable paper bag is made of paper that has a basis weight ranging from 80 to 100 grams per square meter.

[0014] In accordance with another embodiment of the present invention, the foldable paper bag is crafted from Kraft paper.

[0015] In accordance with another embodiment of the present invention, the perforated panel has a hole form between the perforations and the opening of the foldable paper bag, and the hole is aligned with portion of the perforations that is vertically arranged when the foldable paper bag is in an upright position.

[0016] In accordance with another embodiment of the present invention, the perforated panel has a hole, and the hole is located at the position where the first, second, and third straight lines meet.

[0017] In accordance with another embodiment of the present invention, the perforated panel has a hole, and the hole is located at the junction of the vertically flipped "Y" shaped.

[0018] In accordance with another embodiment of the present invention, the perforated panel has a hole, and the hole is located at the position where the front, the back, and the lower sections meet.

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] Embodiments of the invention are described in more details hereinafter with reference to the drawings, in which:

[0020] FIG. **1** depicts a perspective view of a foldable paper bag in accordance with an embodiment of the present invention;

[0021] FIG. **2** depicts a front view of the foldable paper bag in accordance with the embodiment of the present invention;

[0022] FIG. **3** depicts a rear view of the foldable paper bag in accordance with the embodiment of the present invention;

[0023] FIG. **4** depicts a left-side view of the foldable paper bag in accordance with the embodiment of the present invention;

[0024] FIG. **5** depicts a right-side view of the foldable paper bag in accordance with the embodiment of the present invention;

[0025] FIG. **6** depicts a bottom view of the foldable paper bag in accordance with the embodiment of the present invention;

[0026] FIG. **7** depicts a top view of the foldable paper bag in accordance with the embodiment of the present invention;

[0027] FIG. **8** depicts a schematic view of a die-cut of a foldable paper bag in accordance with another embodiment of the present invention;

[0028] FIG. **9** depicts a perspective view of a foldable paper bag in accordance with another embodiment of the present invention; and

[0029] FIG. **10** depicts a perspective view of a foldable paper bag in accordance with another embodiment of the present invention.

DETAILED DESCRIPTION

[0030] In the following description, details of the present invention are set forth as preferred embodiments. It will be apparent to those skilled in the art that modifications, including additions and/or substitutions may be made without departing from the scope and spirit of the invention. Specific details may be omitted so as not to obscure the invention; however, the disclosure is written to enable one skilled in the art to practice the teachings herein without undue experimentation.

[0031] FIG. **1** is a perspective view of a foldable paper bag of an embodiment. In this embodiment, the foldable paper bag **1** has two flat panels **10**, **12**, two perforated panels **11**, **13**, and a bottom panel **14**. The perforated panel **11** connects the flat panels **10**, **12** at one edge, and the perforated panel **13** connects the flat panels **10**, **12** at the other parallel edge. The bottom panel **14** connect the perforated panels **11**, **13**, and the flat panels **10**, **12**. The flat panels **10**, **12** are parallel, ant the perforated panels **11**, **13** are parallel. The flat panels **10**, **12**, the perforated panels **11**, **13** and the bottom panel form a tubular cavity.

[0032] The perforated panel **11** has a plurality of perforations **114**. To be specific, the perforated panel has the first series of perforations **114**A, the second series of perforations **114**B, and the third series of perforations **114**C. The first series of perforations **114**A are arranged along a first straight line, which is parallel with a normal of a surface of the bottom panel **14** in a carrying mode. In the carrying mode, the foldable paper bag **1** is opened and in an upright position.

[0033] The second series of perforations **114**B are arranged in a second straight line, and the third series of perforations **114**C are arranged along a third straight line. An end of the first straight line is adjacent to an opening **15** of the foldable paper bag **1**, and the other end of the first straight line is adjacent to an end of the second straight line and an end of the third straight line. In other words, the first, the second and the third straight line that are defined by the locations of these perforations **114** substantially meet at a lower part of the perforated panel **11**.

[0034] Another end of the second straight line, which is defined by the second series of the perforations **114**B, is connected to a side or edge connecting the perforated panel **11** and the flat panel **10**. Another end of the third straight line, which is defined by the third series of the perforations **114**C, is connected to another side or edge connecting the perforated panel **11** and the flat panel **12**.

[0035] In this embodiment, the perforated panel **11** has perforations **114** arranged in a vertically flipped "Y" shape. When the foldable paper bag **1** is in the upright position, every arm of the vertically flipped "Y" shape is connected to one of the vertical edges of the perforated panel **11**, and the leg of the vertically flipped "Y" shape is vertically extend towards the opening **15** of the foldable paper bag.

[0036] In this embodiment, the perforations **114** on perforated panel **11** divide the panel **11** into a front section **110**, a back section **111**, and a lower section **112**. When the foldable paper bag **1** is in the upright position, the front section **110** is connected to the front flat panel **10**, and the back portion **111** is connected to the back flat panel **12**, and the lower section **112** is connected to the bottom panel **14**.

[0037] The perforated panel **11** of this embodiment has a crease **113**. The crease **113** extends vertically when the foldable paper bag is in an upright position. The crease **113** and the portion of the perforations (i.e., the first series of perforations **114**A) that form a vertical straight line keep a gap in-between. The first straight line defined by the first series of perforations **114**A do not intersect with the crease **113**.

[0038] Moreover, the position or a structure on the perforated panel **11** where the first, second, and third straight lines meet also keep a distance from the crease **113**.

[0039] The perforated panel **11** of this embodiment has a notch **115** on the edge that form the opening **15** of the foldable paper bag **1**. The notch **115** is align with a portion of the perforations (i.e., the first series of perforations **114**A) that is vertically arranged when the foldable paper bag **1** is in an upright position.

[0040] The notch **115** is a V-shape cut in the edge of the perforated panel **11**. To be specific, on the edge of the perforated panel **11**, there is a distinctive V-shaped notch **115**. This small incision creates a symmetrical indentation resembling the letter "V", characterized by two converging lines that meet at a point, forming an acute angle pointing towards the perforations **114**A, to function as a tear starter properly.

[0041] In some embodiments, the foldable paper bag **1** is made of paper, and the orientation of the paper is grain long. The fibers of the paper run parallel to the longer dimension of the perforated panels **11**, **13** and flat panels **10**, **12**.

[0042] In some embodiments, the perforated panel **11** has a hole **116** form on the vertical edge, and the hole **116** is aligned with portion of the perforations (i.e., the second series of the perforations **114**B) that is not vertically arranged when the foldable paper bag **1** is in an upright position. The perforated panel **11** has a hole **117** form on the other vertical edge, and the hole **117** is aligned with portion of the perforations (i.e., the second series of the perforations **114**C) that is not vertically arranged when the foldable paper bag **1** is in an upright position.

[0043] In some embodiments, the hole **116** has a semicircle shape on the perforated panel **11**, which abuts another semicircle shape hole formed on the flat panel **10**. In a die-cut of the foldable paper bag **1**, these two holes form a complete circle, which improve the efficiency of the manufacture of the foldable paper bag **1** easier, and the hole **116** can stop the tearing when user is tearing the perforated panel **11** along the second series of the perforations **114**B.

[0044] In some embodiments, the foldable paper bag **1** is made of paper that has a basis weight that is 80 grams per square meter. In some other embodiments, the paper has a basis weight ranges from 80 to 100 grams per square meter.

[0045] In some embodiments, the foldable paper bag **1** is made of Kraft paper.

[0046] In some embodiments, the perforated panel **11** has a hole **118** form between the perforations **114** and the opening **15**. The hole **118** is aligned with portion of the perforations (i.e., the first series of the perforations **114**A) that is vertically arranged when the foldable paper bag is in an upright position. To be specific, the hole **118** is located between at an end of the first series of the perforations **114**A and the notch **115**.

[0047] In some embodiments, the perforated panel **11** has a hole **119**, and the hole **119** is located at the position where the first, second, and third straight lines meet. In other words, the first, the second, and the third series of the perforation **114**A, **114**B, **114**C extend towards the hole **119**.

[0048] In this embodiment, the hole **119** is in the shape of a quadrilateral, where the top angle is aligned with the first series of the perforations **114**A. One of the side angles is aligned with the second series of the perforations **114**B, and the remaining side angle is aligned with the third series of the perforations **114**C. When a user tears the perforated panel **11** from an end of the first series of the perforations **114**A, the hole **119** can change the direction of tearing easily.

[0049] In other words, when the perforations **114** define the vertically flipped "Y" shape, the hole **119** is located at the junction of the vertically flipped "Y" shaped.

[0050] In other words, the hole **119** is locate at the position where the front, the back, and the lower sections **110**, **111**, **112** meet.

[0051] In the embodiments above, the perforations are a plurality of small hole in the surface of the foldable paper bag **1**. The upright position of the foldable paper bag **1** means the foldable paper bag **1** is standing up-straight while the opening **15** is facing upward, and the bottom panel **14** serves as a base touching the floor or table where the foldable paper bag **1** is disposed.

[0052] In the embodiments above, the perforations **114** are configured to be tear along its arrangement, and the straight lines defined by these perforations **114** intersect with the crease no more than once on every perforated panel.

[0053] FIG. **9** show one of the states when the perforations are torn, and the foldable paper bag **1** can be opened. The flat panels **10**, **12** can be in-folded from the end near the opening **15** of the foldable paper bag **1**. FIG. **10** show another state when the perforations are torn and the foldable paper bag **1** is unfolded. The foldable paper bag offers a broad and suitable surface for users to place their food or groceries, and a compartment is provided to contain any sauce, fries, fingerfood, or granular materials such as salts or sugar.

[0054] While the present disclosure has been described and illustrated with reference to specific embodiments thereof, these descriptions and illustrations are not limiting. The illustrations may not necessarily be drawn to scale. There may be distinctions between the illustrations in the present disclosure and the actual apparatus due to manufacturing processes and tolerances. There may be other embodiments of the present disclosure which are not specifically illustrated. Modifications may be made to adapt a particular situation, material, composition of matter, method, or process to the objective and scope of the present disclosure. All such modifications are intended to be within the scope of the claims appended hereto. While the methods disclosed herein have been described with reference to particular operations performed in a particular order, it will be understood that these operations may be combined, sub-divided, or re-ordered to form an equivalent method without departing from the teachings of the present disclosure. Accordingly, unless specifically indicated herein, the order and grouping of the operations are not limitations.

What is claimed is:

1. A foldable paper bag, comprising:

two flat panels being parallel;

two perforated panels being parallel; and

a bottom panel;

wherein each of the perforated panels connects the flat panels, and the bottom panel connects the flat panels and the perforated panels;

wherein each of the perforated panels has a first series of perforations, a second series of perforations, and a third series of perforations;

wherein the first series of perforations are arranged along a first straight line, which is parallel with a normal of a surface of the bottom panel in a carrying mode;

wherein the second series of perforations are arranged along a second straight line;

wherein the third series of perforations are arranged along a third straight line; and

wherein an end of the first straight line is adjacent to an opening of the foldable paper bag, and the other end of the first straight line is adjacent to an end of the second straight line and an end of the third straight line, and another end of the second straight line is connected to a side connecting the perforated panel and one of the flat panels, and another end of the third straight line is connected to another side connecting the perforated panel and another flat panel.

2. The foldable paper bag of claim **1**, wherein every perforated panel has a first hole, and the first hole is located at the position where the first, second, and third straight lines meet.

3. The foldable paper bag of claim **1**, wherein every perforated panel has a crease, and the crease extends vertically when the foldable paper bag is in an upright position, and the crease and the portion of the perforations that form a vertical straight line keep a gap in-between.

4. The foldable paper bag of claim **1**, wherein every perforated panel has a notch on the edge that form the opening of the foldable paper bag, and the notch is align with a portion of the perforations that is vertically arranged when the foldable paper bag is in an upright position.

5. The foldable paper bag of claim **1**, wherein every perforated panel has a second first hole form on the vertical edge, and the second first hole is aligned with portion of the perforations that is not vertically arranged when the foldable paper bag is in an upright position.

6. The foldable paper bag of claim **1**, wherein every perforated panel has a third first hole form between the perforations and the opening of the foldable paper bag, and the third first hole is aligned with portion of the perforations that is vertically arranged when the foldable paper bag is in an upright position.

7. A foldable paper bag, comprising:

two flat panels being parallel;

two perforated panels being parallel;

a bottom panel;

wherein every perforated panel has perforations arranged in a vertically flipped "Y" shape having a leg and two arms;

wherein when the foldable paper bag is in an upright position, every arm of the vertically flipped "Y" shape is connected to one of the vertical edges of the perforated panel, and the leg of the vertically flipped "Y" shape is vertically extend towards an opening of the foldable paper bag.

8. The foldable paper bag of claim **7**, wherein the perforated panel has a first hole, and the first hole is located at the junction of the vertically flipped "Y" shape.

9. The foldable paper bag of claim **7**, wherein every perforated panel has a crease, and the crease extends vertically when the foldable paper bag is in an upright position, and the crease and the portion of the perforations that form a vertical straight line keep a gap in-between.

10. The foldable paper bag of claim **7**, wherein every perforated panel has a notch on the edge that form the opening of the foldable paper bag, and the notch is align with a portion of the perforations that is vertically arranged when the foldable paper bag is in an upright position.

11. The foldable paper bag of claim **7**, wherein every perforated panel has a second first hole form on the vertical edge, and the second first hole is aligned with portion of the perforations that is not vertically arranged when the foldable paper bag is in an upright position.

12. The foldable paper bag of claim **7**, wherein every perforated panel has a third first hole form between the perforations and the opening of the foldable paper bag, and the third first hole is aligned with portion of the perforations that is vertically arranged when the foldable paper bag is in an upright position.

13. A foldable paper bag, comprising:

two flat panels being parallel;

two perforated panels being parallel;

a bottom panel;

wherein every perforated panel has perforations that divided the perforated panel into a front section, a back section, and a lower section; and

wherein when the foldable paper bag is in an upright position, the front section is connected to the front flat panel, and the back section is connected to the back flat panel, and the lower section and a bottom panel of the foldable paper bag are connected.

14. The foldable paper bag of claim **13**, wherein the perforated panel has a first hole, and the first hole is located at the position where the front, the back, and the lower sections meet.

15. The foldable paper bag of claim **13**, wherein every perforated panel has a crease, and the crease extends vertically when the foldable paper bag is in an upright position, and the crease and the portion of the perforations that form a vertical straight line keep a gap in-between.

16. The foldable paper bag of claim **13**, wherein every perforated panel has a notch on the edge that form the opening of the foldable paper bag, and the notch is align with a portion of the perforations that is vertically arranged when the foldable paper bag is in an upright position.

17. The foldable paper bag of claim **13**, wherein every perforated panel has a second first hole form on the vertical edge, and the second first hole is aligned with portion of the perforations that is not vertically arranged when the foldable paper bag is in an upright position.

18. The foldable paper bag of claim **13**, wherein every perforated panel has a third first hole form between the perforations and the opening of the foldable paper bag, and the third first hole is aligned with portion of the perforations that is vertically arranged when the foldable paper bag is in an upright position.

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